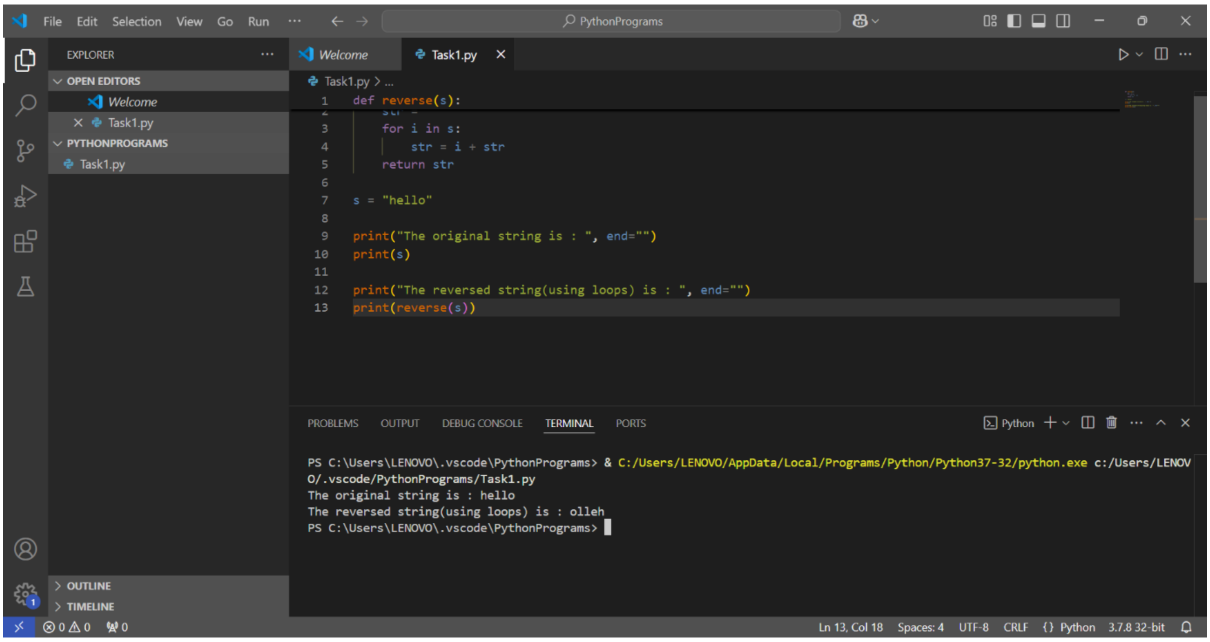
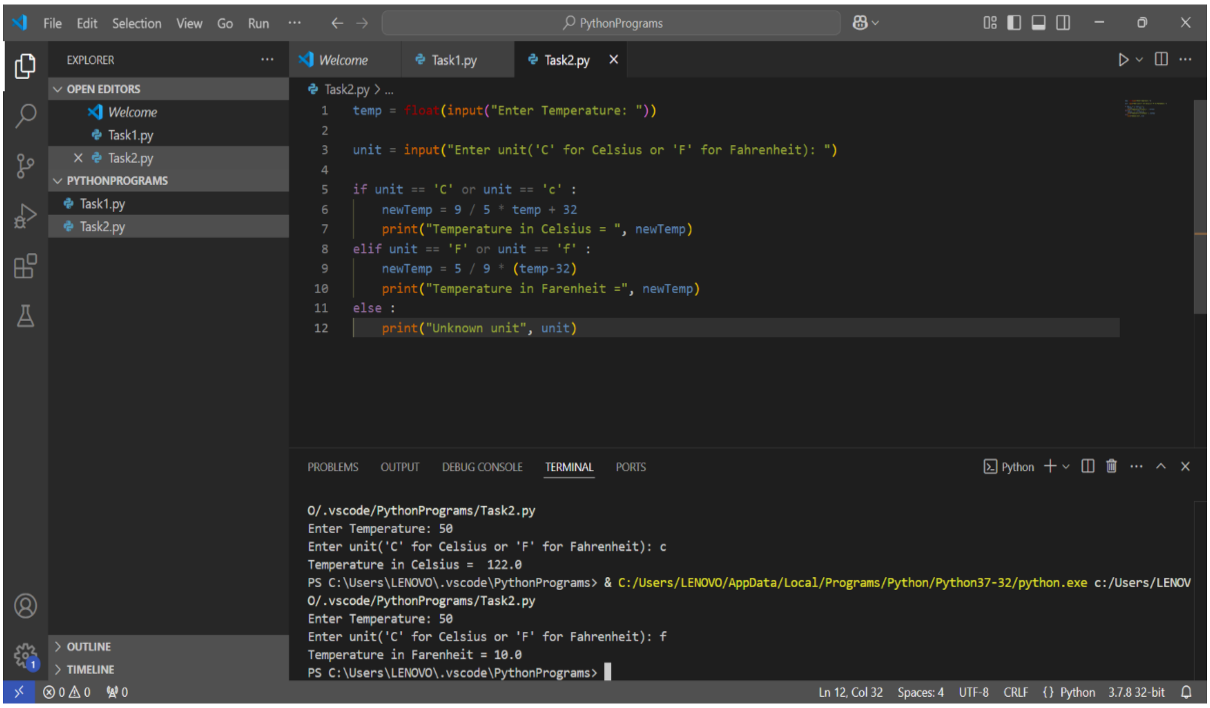
COGNIFY TECHNOLOGIES – PYTHON DEVELOPMENT INTERNSHIP

LEVEL 1

Task 1: **String Reversal**: Create a Python function that takes a string as input and returns the reverse of that string. For example, if the input is "hello," the function should return "olleh."



Task 2: **Temperature Conversion** :Create a Python program that converts temperatures between Celsius and Fahrenheit. Prompt the user to enter a temperature value and the unit of measurement, and then display the converted temperature.



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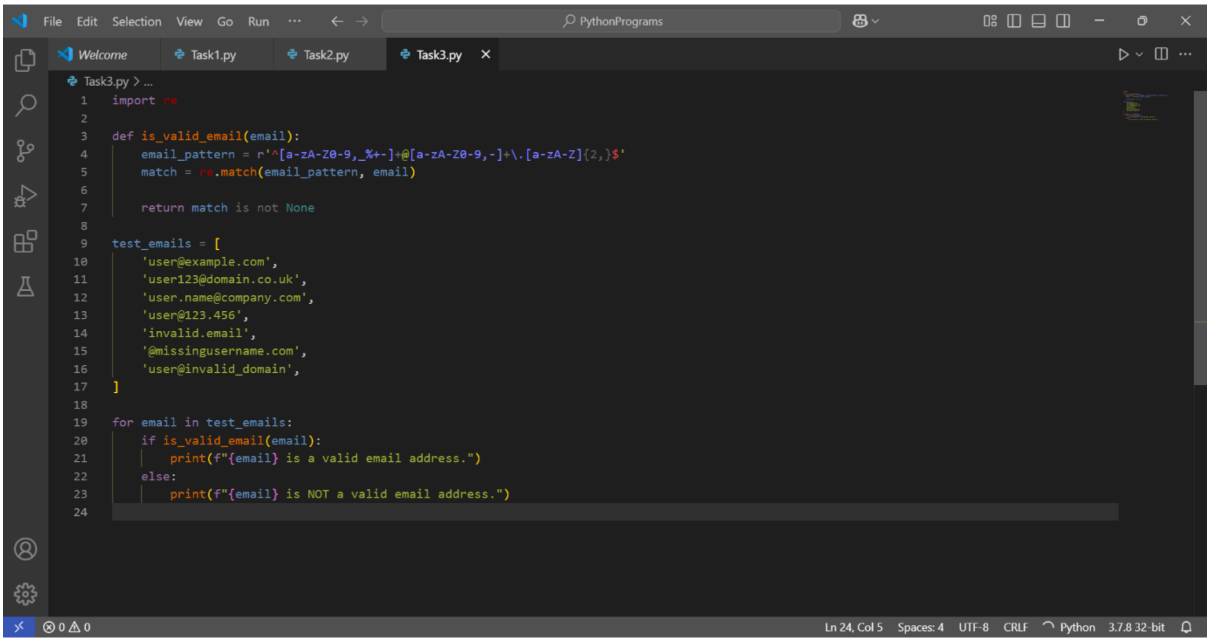
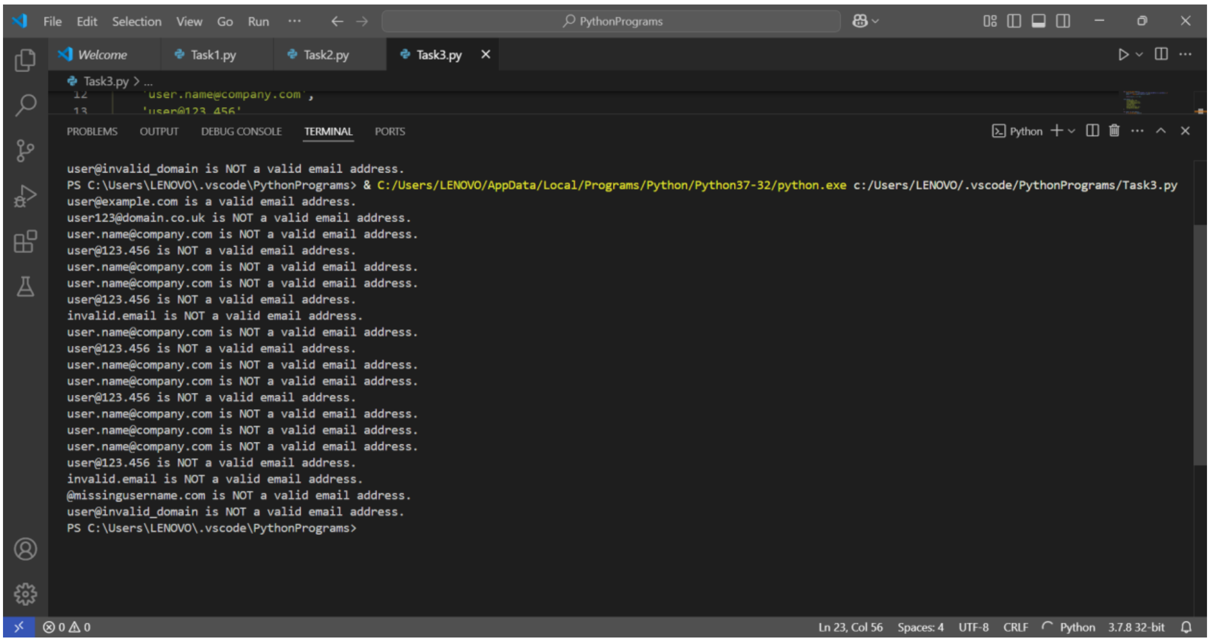
Task 3: **Email Validator**: Develop a Python function that validates whether a given string is a valid email address. Implement checks for the format, including the presence of an "@" symbol and a domain name.



Output



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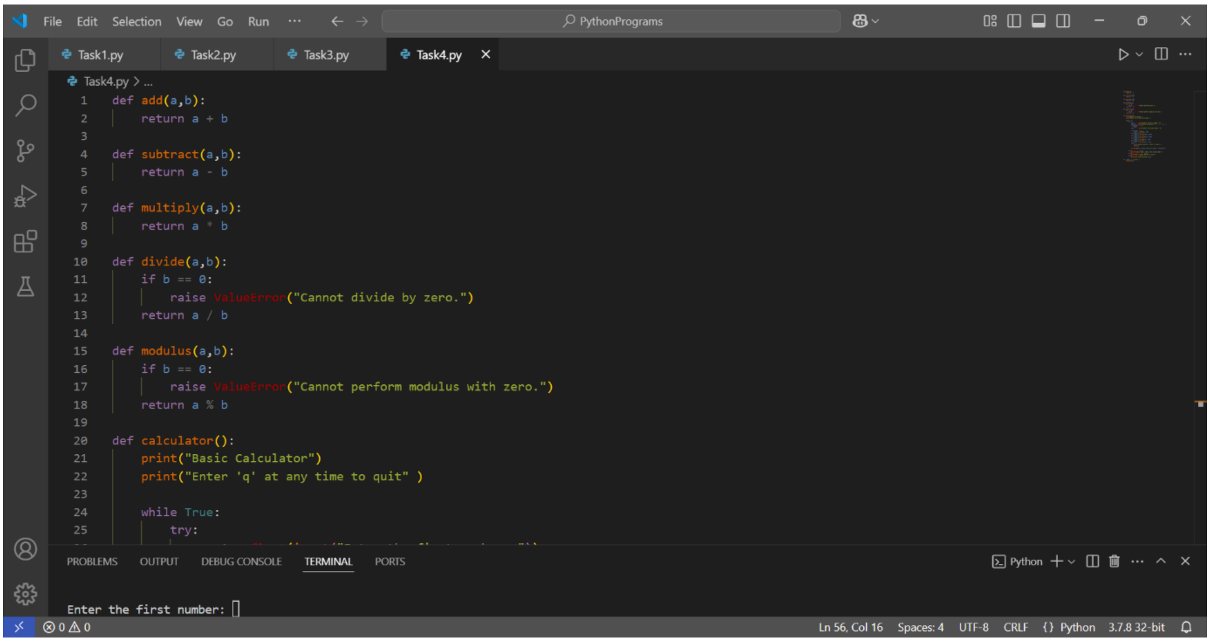
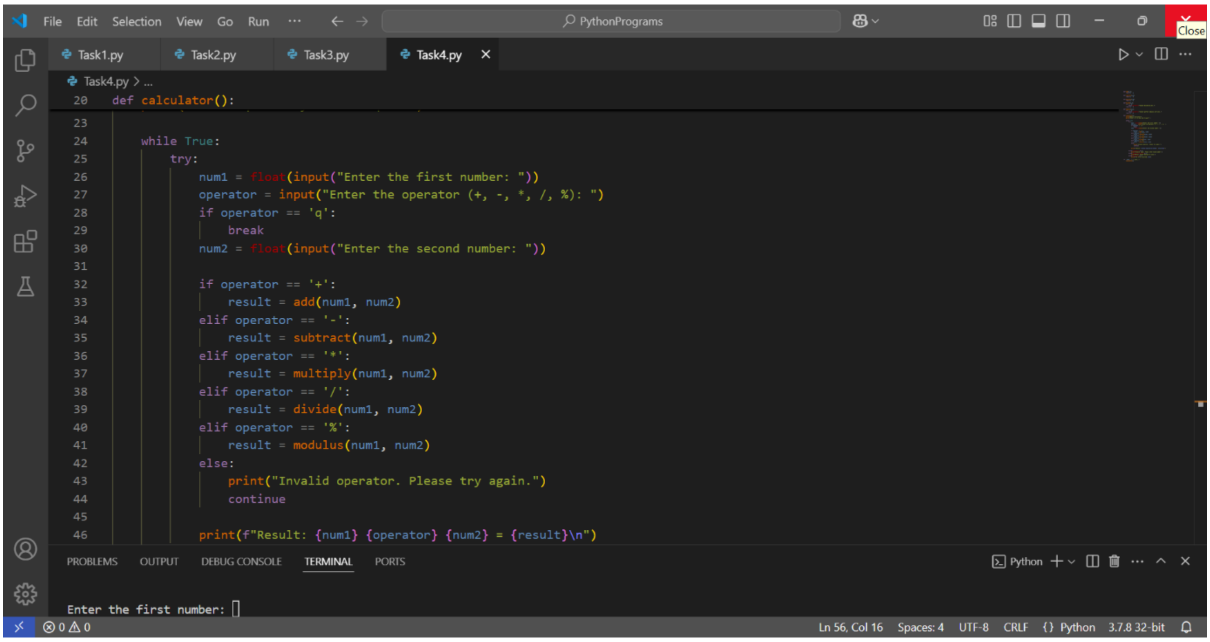
Task 4: **Calculator Program:** Create a Python program that acts as a basic calculator. It should prompt the user to enter two numbers and an operator (+, -, *, /, %), and then display the result of the operation.





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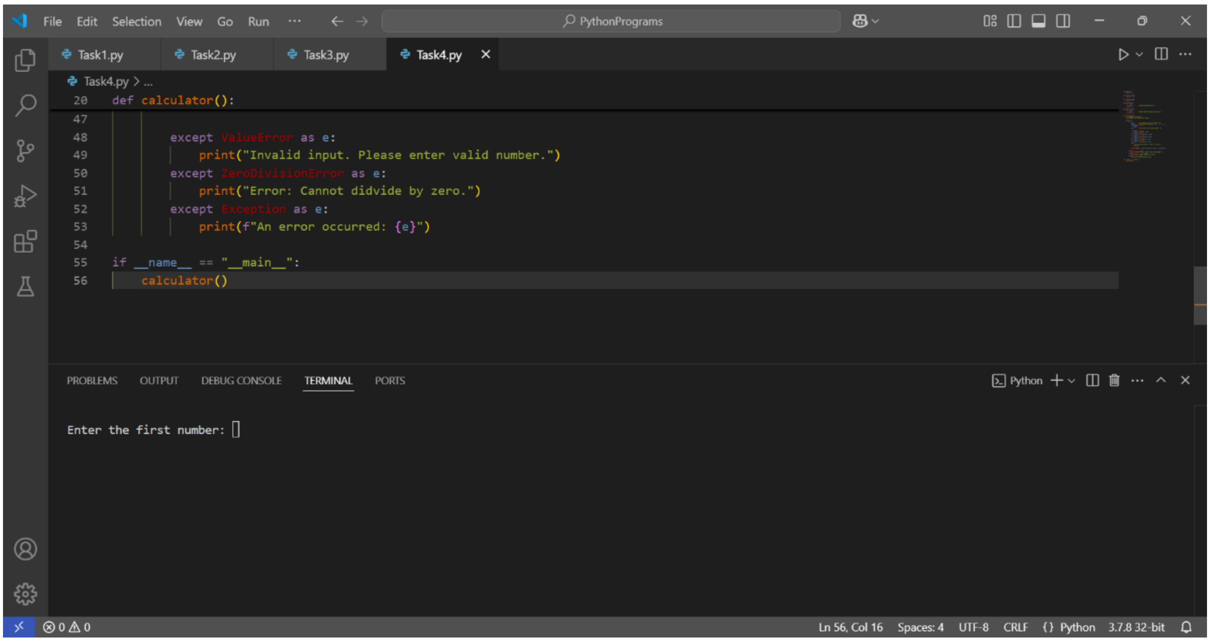
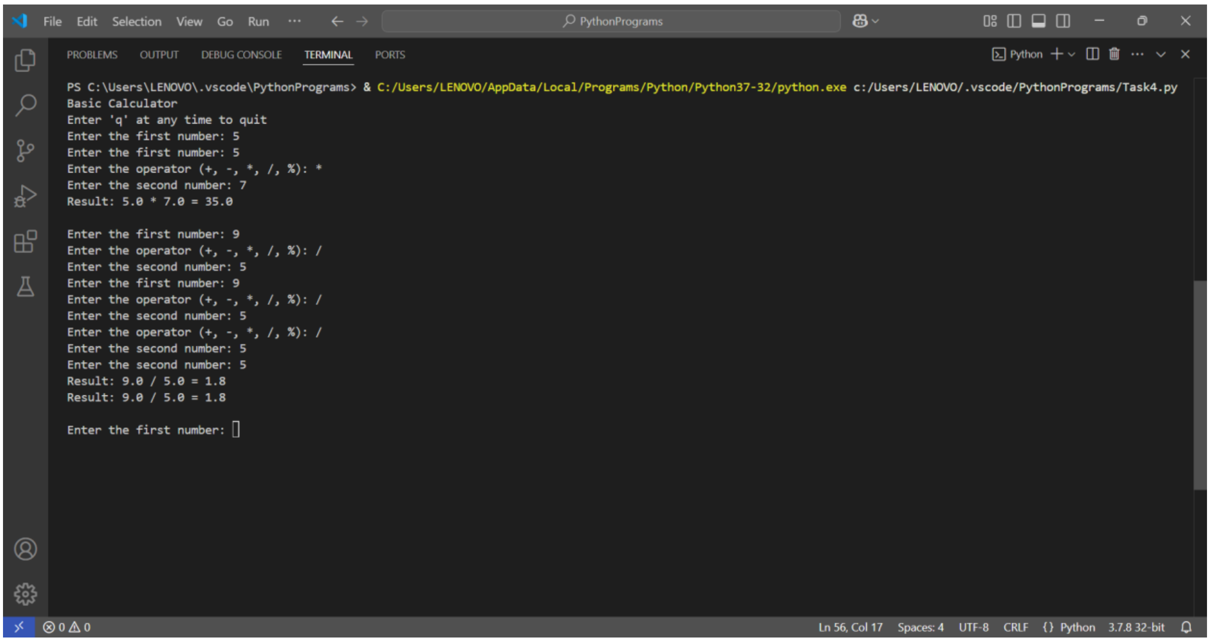


Output



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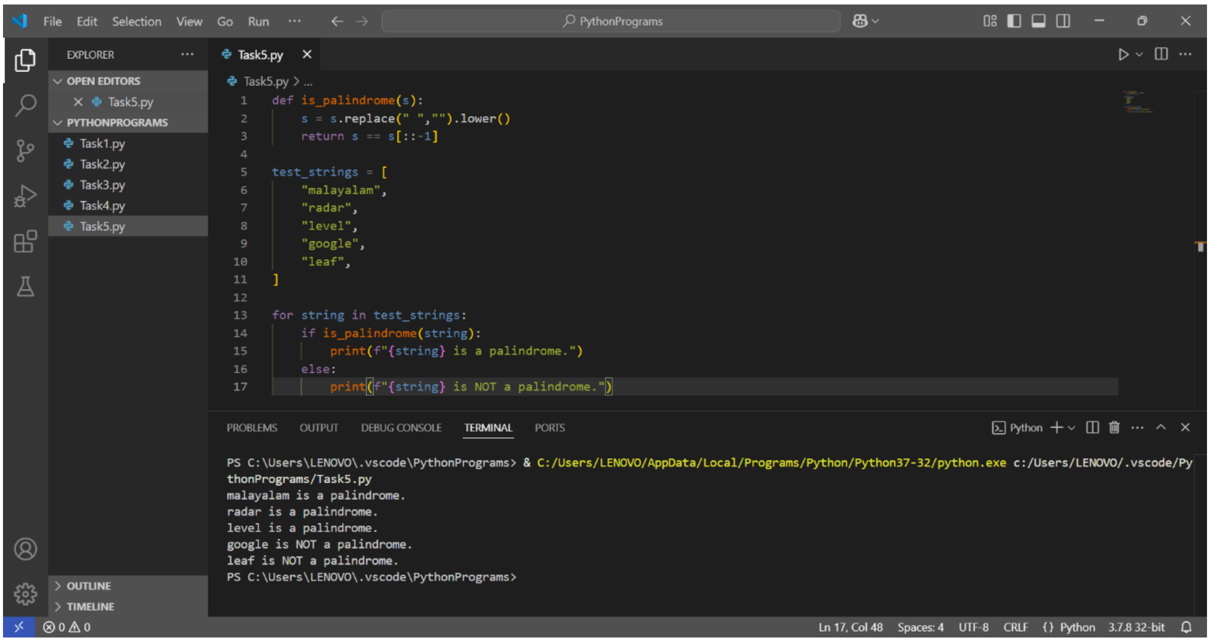
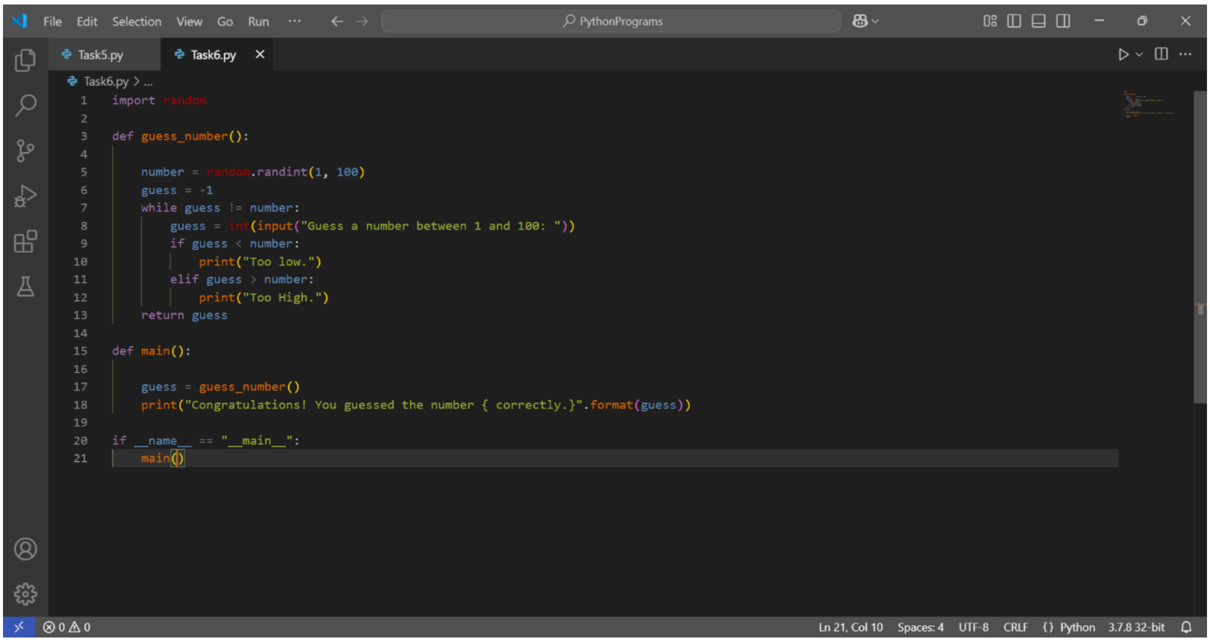
Task 5: **Palindrome Checker** : Write a Python function that checks whether a given string is a palindrome. A palindrome is a word, phrase, or sequence that reads the same backward as forward (e.g., "madam" or "racecar").



LEVEL 2

Task1:**Guessing Game** : Write a Python program that generates a random number between 1 and 100. The user should then try to guess the number. The program should provide hints such as "too high" or "too low" until the correct number is guessed.



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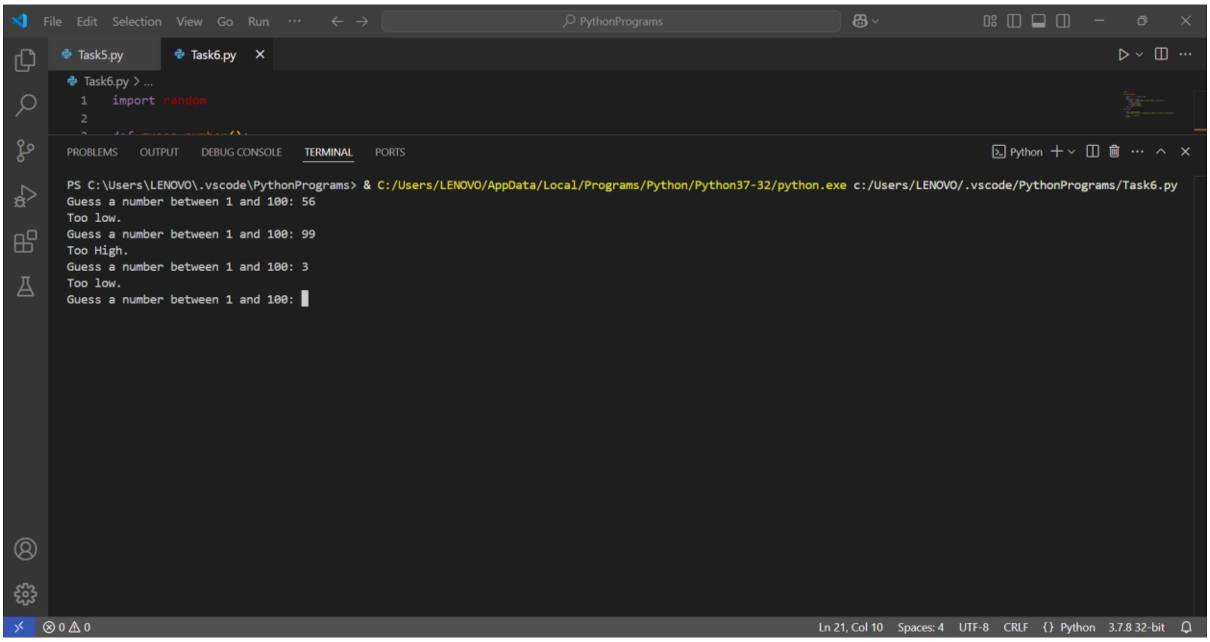

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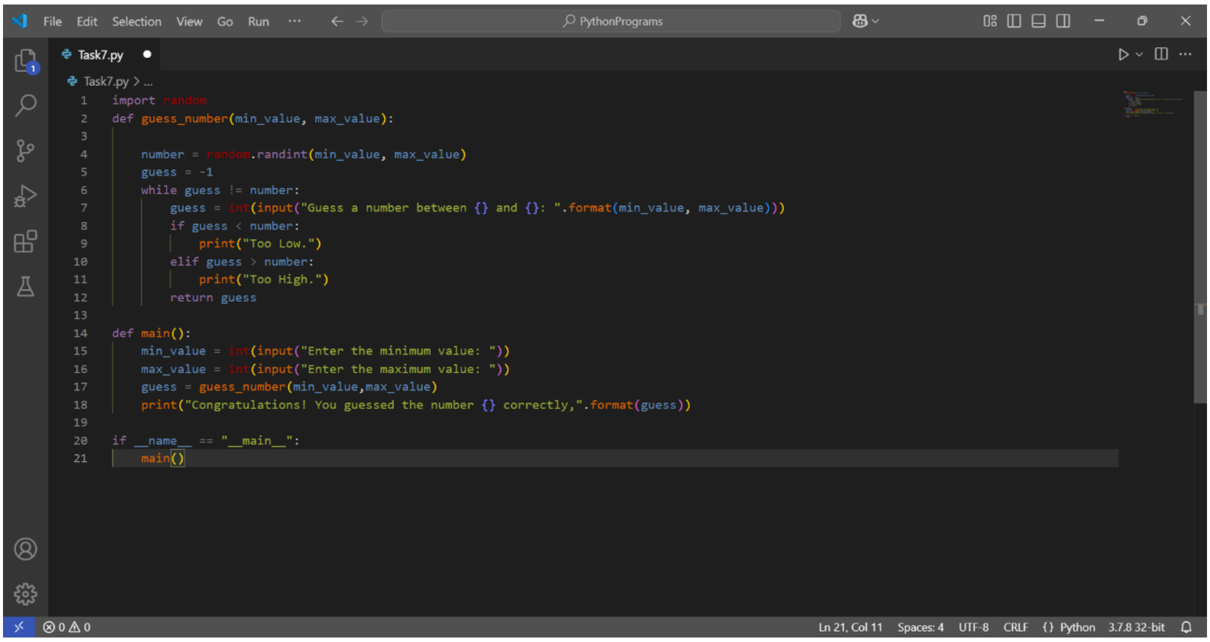
Task 2: **Number Guesser** : Create a number guessing game where the program generates a random number between a specified range, and the user tries to guess it. Provide feedback to the user if their guess is too high or too low.



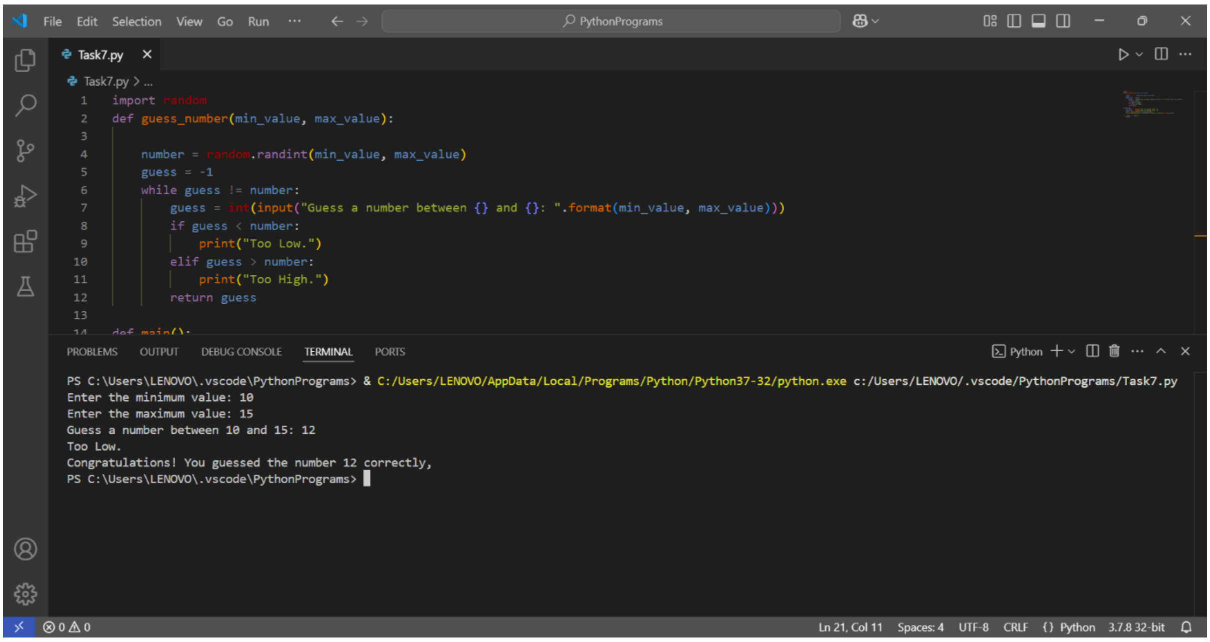
```python
import random
def guess_number(min_value, max_value):

    number = random.randint(min_value, max_value)
    guess = -1
    while guess != number:
        guess = int(input("Guess a number between {} and {}: ".format(min_value, max_value)))
        if guess < number:
            print("Too Low.")
        elif guess > number:
            print("Too High.")
        return guess

def main():
    min_value = int(input("Enter the minimum value: "))
    max_value = int(input("Enter the maximum value: "))
    guess = guess_number(min_value,max_value)
    print("Congratulations! You guessed the number {} correctly,".format(guess))

if __name__ == "__main__":
    main()
```

Output



```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task7.py
Enter the minimum value: 10
Enter the maximum value: 15
Guess a number between 10 and 15: 12
Too Low.
Congratulations! You guessed the number 12 correctly,
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

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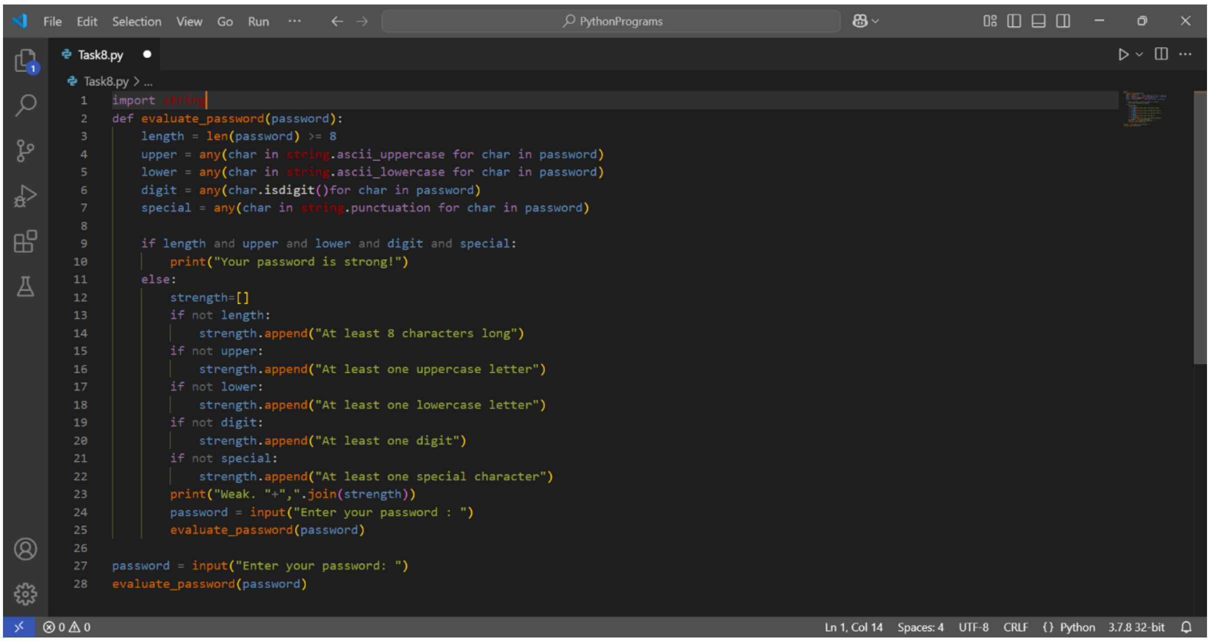
Task 3**: Password Strength**: Checker Create a Python function that evaluates the strength of a password entered by the user. Implement checks for factors such as length, presence of uppercase and lowercase letters, digits, and special characters.

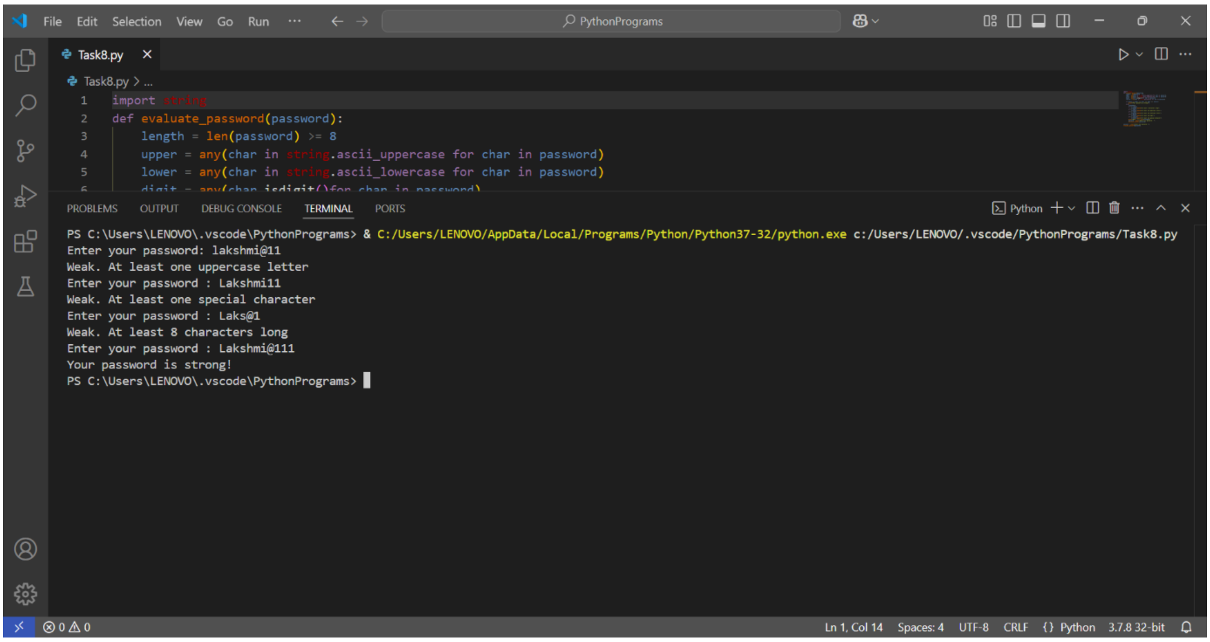


```python
import string
def evaluate_password(password):
    length = len(password) >= 8
    upper = any(char in string.ascii_uppercase for char in password)
    lower = any(char in string.ascii_lowercase for char in password)
    digit = any(char.isdigit()for char in password)
    special = any(char in string.punctuation for char in password)

    if length and upper and lower and digit and special:
        print("Your password is strong!")
    else:
        strength=[]
        if not length:
            strength.append("At least 8 characters long")
        if not upper:
            strength.append("At least one uppercase letter")
        if not lower:
            strength.append("At least one lowercase letter")
        if not digit:
            strength.append("At least one digit")
        if not special:
            strength.append("At least one special character")
        print("Weak. "+",".join(strength))
        password = input("Enter your password : ")
        evaluate_password(password)

password = input("Enter your password: ")
evaluate_password(password)
```

Output



```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task8.py
Enter your password: lakshmi@11
Weak. At least one uppercase letter
Enter your password : Lakshmi11
Weak. At least one special character
Enter your password : Laks@1
Weak. At least 8 characters long
Enter your password : Lakshmi@111
Your password is strong!
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```

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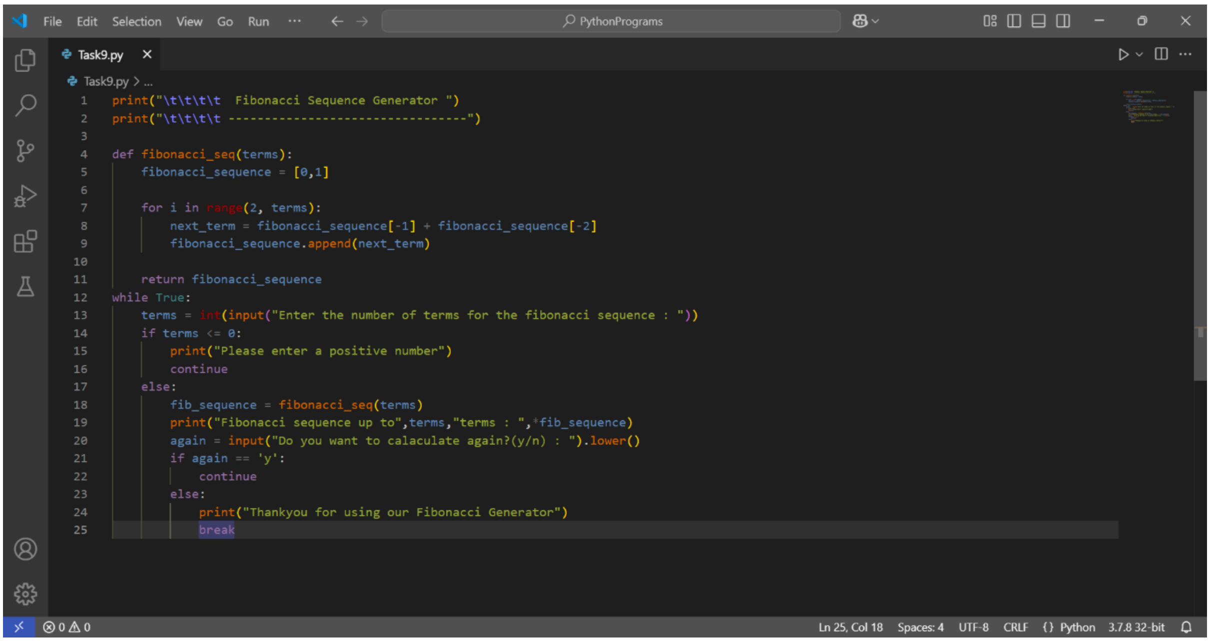
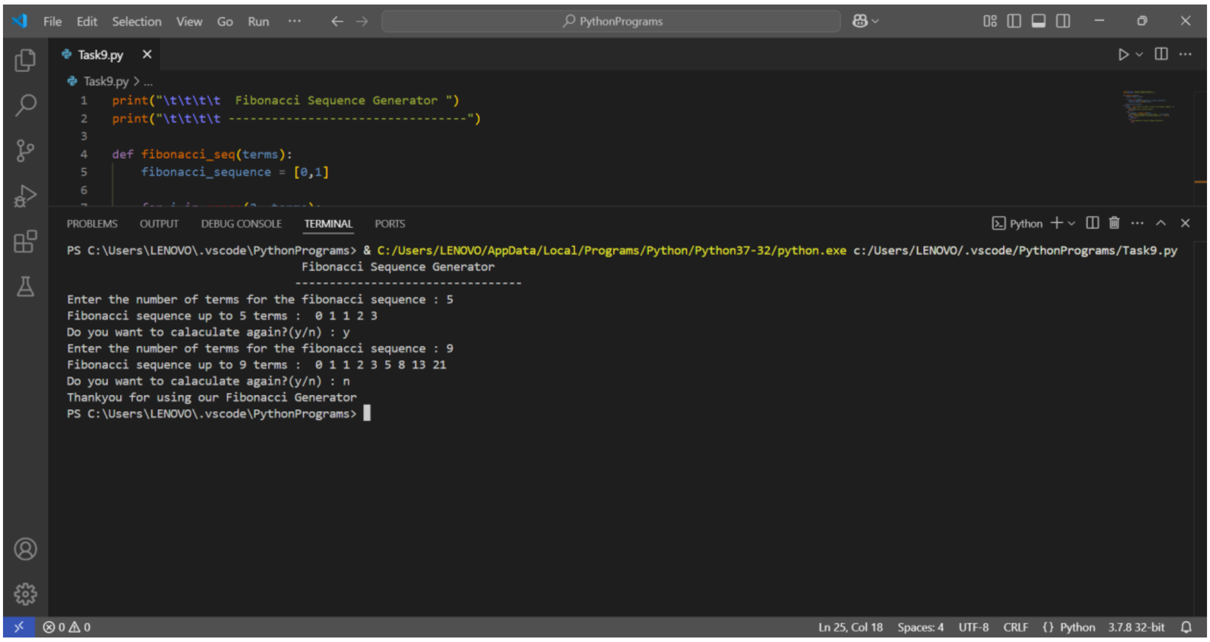
Task 4: **Fibonacci Sequence**: Write a Python function that generates the Fibonacci sequence up to a given number of terms. The function should take an integer input from the user and display the Fibonacci sequence up to that number of terms.





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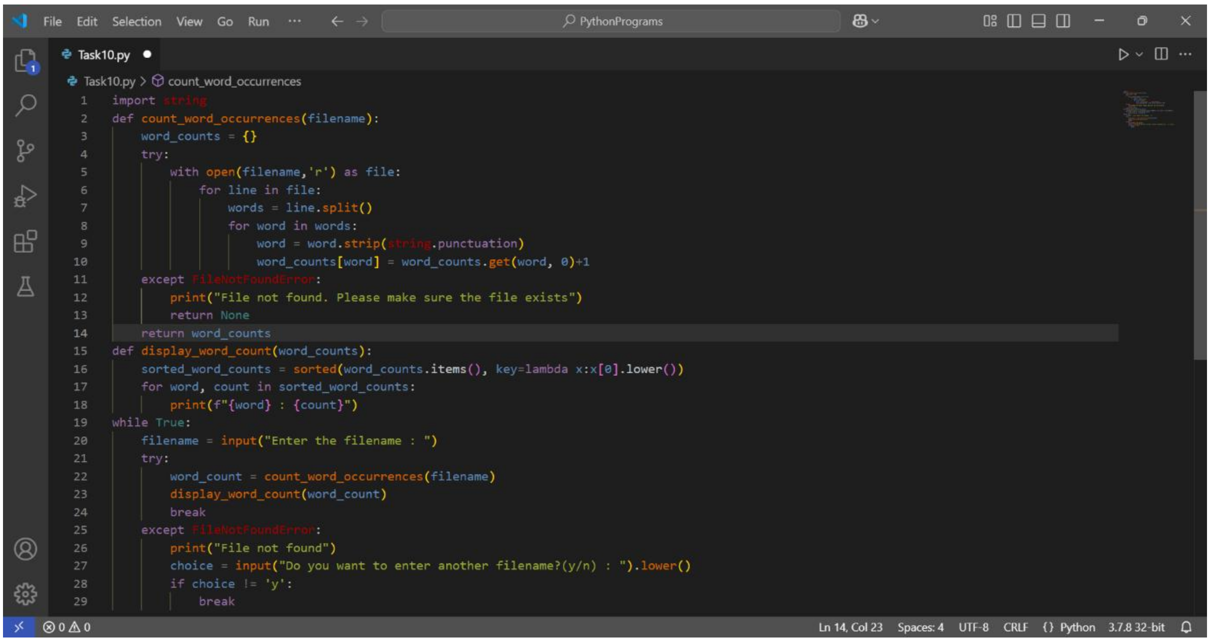
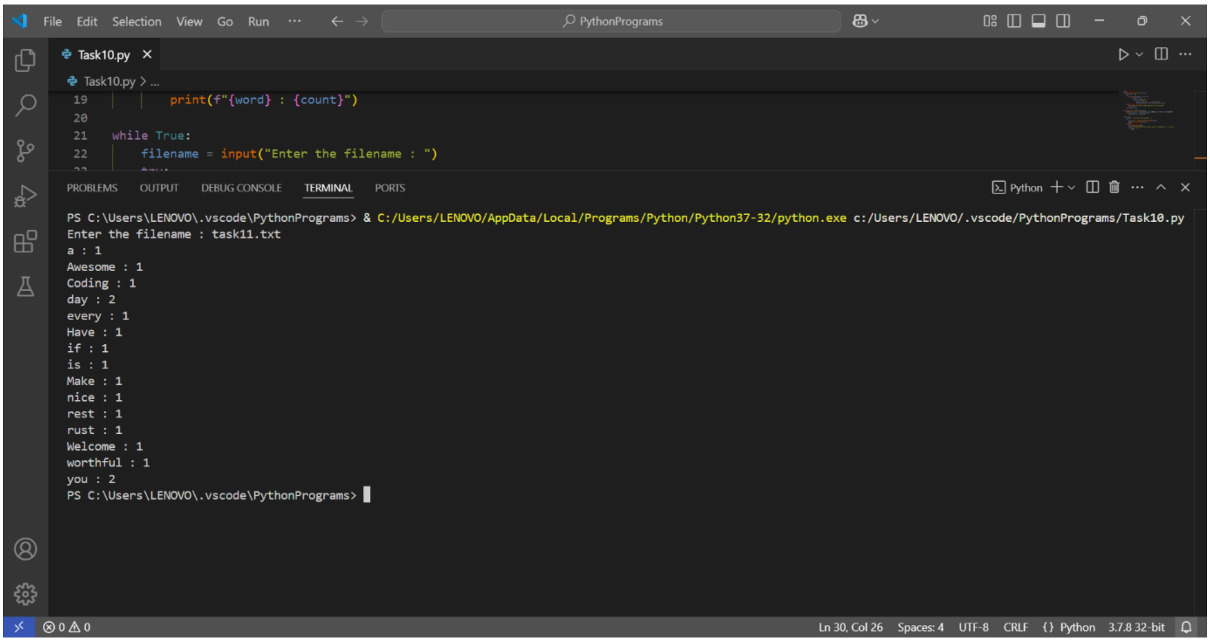
Task 5: File Manipulation Write a Python program that reads a text file and counts the occurrences of each word in the file. Display the results in alphabetical order along with their respective counts.



```python
import string
def count_word_occurrences(filename):
    word_counts = {}
    try:
        with open(filename,'r') as file:
            for line in file:
                words = line.split()
                for word in words:
                    word = word.strip(string.punctuation)
                    word_counts[word] = word_counts.get(word, 0)+1
    except FileNotFoundError:
        print("File not found. Please make sure the file exists")
        return None
    return word_counts
def display_word_count(word_counts):
    sorted_word_counts = sorted(word_counts.items(), key=lambda x:x[0].lower())
    for word, count in sorted_word_counts:
        print(f"{word} : {count}")
while True:
    filename = input("Enter the filename : ")
    try:
        word_count = count_word_occurrences(filename)
        display_word_count(word_count)
        break
    except FileNotFoundError:
        print("File not found")
        choice = input("Do you want to enter another filename?(y/n) : ").lower()
        if choice != 'y':
            break
```

Output



```
PS C:\Users\LENOVO\.vscode\PythonPrograms> & C:/Users/LENOVO/AppData/Local/Programs/Python/Python37-32/python.exe c:/Users/LENOVO/.vscode/PythonPrograms/Task10.py
Enter the filename : task11.txt
a : 1
Awesome : 1
Coding : 1
day : 2
every : 1
Have : 1
if : 1
is : 1
Make : 1
nice : 1
rest : 1
rust : 1
Welcome : 1
worthful : 1
you : 2
PS C:\Users\LENOVO\.vscode\PythonPrograms>
```